COUNTRY PROFILES

# Nuclear Power in Belgium

UPDATED TUESDAY, 9 DECEMBER 2025

- Belgium has three operable nuclear reactors.
- Belgium's first commercial nuclear power reactor began operating in 1974.
- In 2003 a federal law was introduced that required the phase-out of all nuclear electricity generation.
- Plans to close all nuclear power plants by 2025 were delayed in March 2022 by 10 years, with the country's

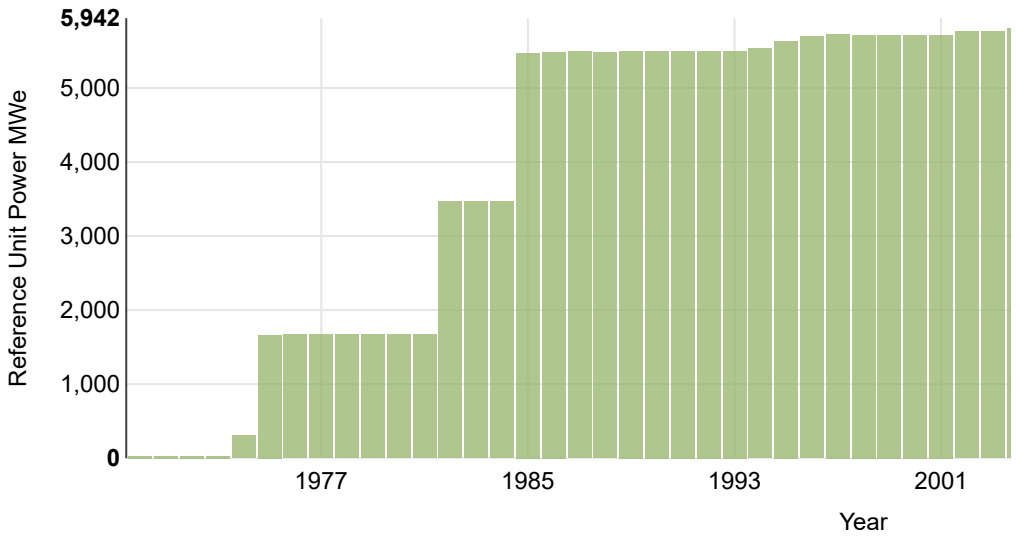
Privacy - Terms

two newest reactors, Doel 4 and Tihange 3, allowed to remain in operation to 2035.

- In May 2025 Belgium's parliament repealed the 2003 phase-out law.
- The government now intends to extend the operating licences of Doel 4 and Tihange 3 to 2045, and construct new reactors.

| 2 | 0 | 6 |
|---|---|---|
| OPERABLE REACTORS | REACTORS UNDER CONSTRUCTION | REACTORS SHUTDOWN |
| 2,056 MWe | 0 MWe | 3,882 MWe |

### Operable nuclear power capacity




## Electricity sector

**Total generation (in 2024):** 620 TWh

**Generation mix:** nuclear 31.3 TWh (41%); wind 14.0 TWh (18%); natural gas 13.4 TWh (18%); solar 8.0 TWh (11%); biofuels & waste 4.7 TWh (6%); coal 2.2 TWh (3%); hydro 1.6 TWh (2%).

**Import/export balance:** 10.3 TWh net import (27.8 TWh imports, 17.4 TWh exports)

**Total consumption:** 86.0 TWh

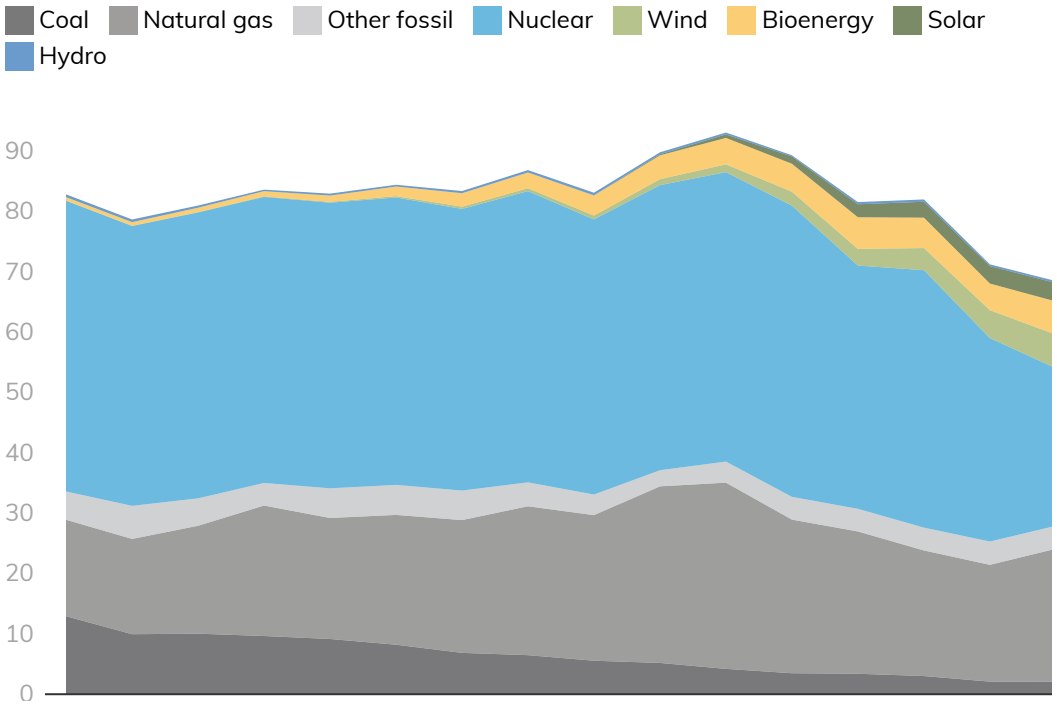
**Per capita consumption:** c. 7200 kWh in 2024

*Source: International Energy Agency, The World Bank. Data for year 2024.*

Belgium's electricity demand been stable over the past 20 years. Per capita electricity consumption in the country reduced from about 8200 kWh in 2000 to about 6800 kWh in 2023.

Under the (now repealed) 2003 law to phase out nuclear, the share of electricity generated by the country's reactors declined from a high of about 60% at the turn of the century to about 40% now.

**Belgium electricity generation by source (TWh)**




Source: EMBER Electricity Data Explorer

Most Belgian electricity is produced by Electrabel, a subsidiary of GDF Suez (now Engie)[a], which also operates all the nuclear plants.

The transmission system operator (TSO) is Elia.

## Nuclear power plants

Belgium has two commercial nuclear power plants with operable reactors. Both power plants are operated by ENGIE Electrabel.

A small prototype reactor, BR-3, was operated by SCK CEN until it was shut down in 1987.

**Operable reactors in Belgium**

| Reactor name | Model | Reactor type | Net capacity (MWe) | Construction start | Grid connection |
|---|---|---|---|---|---|
| Doel 4 | W (3-loop) | PWR | 1000 | 1978-12-01 | 1985-04-08 |
| Tihange 3 | W (3-loop) | PWR | 1020 | 1978-11-01 | 1985-06-15 |

# Nuclear industry development

In 1962, Belgium's first small nuclear power reactor was commissioned at Mol – the 11 MWe prototype BR3 PWR, which was imported from the USA. This was the first pressurized water reactor (PWR) built in Europe.

In 1965, Synatom was formed as a syndicate for design of large nuclear power plants. It is now a subsidiary of Electrabel, which is part of the Energy Europe & International business unit of GDF Suez (now Engie).

In 1966, the Franco-Belgian 305 MWe Chooz A prototype PWR was commissioned just across the border in France, and it was decided to build the Doel 1&2 and Tihange 1 reactors in Belgium, the last being a Franco-Belgian project. These were commissioned in 1974-75 and upgraded in 1985 to improve seismic and other safety. Chooz A was closed in 1991 and is being decommissioned by EDF.

In 1974, a further four reactors were ordered for Doel and Tihange sites. Doel is in Flanders (north) and Tihange is in Wallonia (south). It was decided in 1988 not to proceed with an eighth unit, which was to be a 1400 MWe N4 type at Doel jointly owned by Electricité de France (EDF) and Electrabel.

In addition, Electrabel has a 25% share in output of the two 1500 MWe Chooz B units (effectively adding 750 MWe) and 458 MWe of Tricastin operated by EDF in France. Reciprocally, EDF owns half of Tihange 1, now shutdown. Electrabel formerly had entered into a capacity swap arrangement with E.ON that included E.ON

acquiring drawing rights for 770 MWe of capacity from Doel 1, Doel 2 and Tihange 1; and Electrabel 700 MWe drawing rights from the Krümmel, Gundremmingen and Unterweser nuclear plants in Germany (all now shutdown)[1].

Belgian companies provided about 80% of the systems and equipment for the country's nuclear facilities, though both Westinghouse and Framatome contributed.

## Upgrading and reactor life extension

The steam generators were replaced in Doel 2, 3&4 and Tihange 1, 2&3 between 1993 and 2004; and those of Doel 1 were replaced in 2009. Capacity uprates were also carried out during this period.

In 2009, Electrabel began preparing the safety case for extending the operating lifetimes of Doel 1&2 and Tihange 1 by ten years. In May 2010, Electrabel called for bids to replace the turbines at Doel 1&2 over 2011-15, including new instrumentation. Electrabel applied for a 10-year licence extension for Doel 1&2 in 2012, and the government agreed to this in principle in December 2014 subject to FANC approval (despite the 2012 policy described below). With FANC approval and specific government agreement not forthcoming, Electrabel closed Doel 1 in mid-February 2015 as required by law, and prepared to defuel it. The company said earlier: "The Minister of Energy and Electrabel Group GDF Suez continue the discussions on the lifetime extension of Doel 1&2. Electrabel participates constructively in these discussions while recalling the need to define a clear legal and economic framework to consider investing locally €600 to 700 million needed for the extension of [Doel 1's operating lifetime]."

At the end of February 2015 the company said that it had ordered new fuel for Doel 1 and it could be restarted by the end of the year if agreement could be reached with the government on an operating lifetime extension plan. Doel 2 was scheduled to close on November 30, but with shutdown for two months in 2015 it could then run to March 2016. Negotiations with government included removal of the nuclear generation tax, which cost the company €397 million in 2014.

Legislation to enable 10-year life extensions for Doel 1&2 to 2025 was passed by parliament in June 2015, thus overturning a major provision of the 2003 law. At the end of the following month the government said it had reached agreement with Electrabel on the

financial implications. The company agreed to pay a €20 million per year fee for lifetime extension of Doel 1&2 from 2016, plus €200 million in 2015 and €130 million in 2016 as the federal nuclear power 'contribution' or tax, substantially less than previously intended (and less than the €479 million in 2014). From 2017 a formula was applied, with a minimum of €150 million per year to 2019. The revised tax for all units and the annual fee for Doel 1&2 was signed off at the end of November. Electrabel had reached agreement with FANC on upgrades to the units in September, clearing the way for reactivating the licence. FANC approved restarting the units in December, after priority upgrades had been completed.

Engie said the November 2015 agreement established "a stable legal and economic framework for the future," and that it planned to spend €700 million to upgrade Doel 1&2. This included ultrasound checks on both reactor pressure vessels, due to the concerns with unit 3. They also needed to prove compliance with new seismic standards of 100 Gal peak ground acceleration proposed by WENRA. The IAEA completed a safety aspects of long-term operation (SALTO) review of Doel 1&2, at the invitation of FANC. The IAEA team concluded that the plant had made significant progress on ageing management and preparation for safe long-term operation.

Earlier in January 2014 joint owners EDF and Electrabel agreed to invest €600 million in upgrading and operating lifetime extension for Tihange 1 to 2025, but with the government taking €1.25 billion in resulting windfall profits over ten years.

In March 2017 the European Commission approved under state aid rules the government agreements with Engie, Electrabel, and EDF to spend €1.3 billion to upgrade the three reactors and run them to 2025.

Earlier In 2012 Doel 3 and Tihange 2 were shut down due to concern about flaws in the reactor pressure vessels due to hydrogen flakes from their manufacture. After thorough investigations, Electrabel submitted a report to the Federal Agency for Nuclear Control (FANC) confirming their structural integrity with significant margins. After requirements of FANC were met, it approved restarting the units in May 2013. However, further metallurgical testing led in March 2014 to Electrabel shutting them down again and bringing forward maintenance outages for both plants until uncertainties regarding the effect of neutrons on mechanical strength of the RPV steel were resolved. First-of-its-kind testing procedures were undertaken at the Belgian Nuclear

Research Centre. On the basis of these GdF Suez (now Engie) submitted a report to FANC and its technical subsidiary, Bel V, which was considered by an international expert panel in November 2014. Further tests were ordered, and the results were considered by the reconvened panel in April 2015. Electrabel reported on “tens of thousands of hours of investigative work and more than 1500 material tests” following up the ultrasound results. FANC authorized their restart in November 2015, saying that hydrogen flakes in the walls of the pressure vessels have “no unacceptable impact” on reactor safety. Parent company Engie said that the extended outage of the two units had cost the group some €40 million per month. Both reactors restarted in December.

In February 2023 the Belgium government asked Engie to investigate whether Doel 1&2 and Tihange 1 could operate beyond their 2025 shutdown dates, so that they could be held in reserve for the winter months of 2025-2027.

In December 2023 the government finalized a deal with Engie to implement a ten-year extension of Doel 4 and Tihange 3. In July 2025 FANC gave the go-ahead for the restart of Tihange 3 following maintenance work in preparation for a further ten years of operation.

## Recent nuclear energy policy

In 1999, the Belgian government appointed the AMPERE commission (*Commission d'Analyse des Modes de Production d'Électricité et de Redéploiement des Énergies*) to report on electricity demand and options for meeting it in the 21st century. It also announced that reactor operating lifetimes would be limited to 40 years, and banned further reprocessing. The commission reported in 2000 that nuclear power was important to Belgium and recommended its further development.

However, due to political factors in the governing coalition, the Belgian Senate approved the Federal Act of 31 January 2003, which prohibited the building of new nuclear power plants and limited the operating lifetimes of existing ones to 40 years (to 2014-2025). This could be overridden by a recommendation from the electricity and gas regulator (*Commission de Régulation de l'Électricité et du Gaz*, CREG) if Belgium's security of supply is threatened.

In 2007, the *Commission on Energy 2030*ᶜ energy policy study set up by the government said that a fundamental review of energy

policy was required and in particular that nuclear power should be utilized long-term in order to meet carbon dioxide ($CO_2$) reduction commitments, enhance energy security and maintain economic stability. It also said that the 2003 phase-out decision should be reconsidered as it would double the price of electricity, deny Belgium a cheap way of meeting the country's $CO_2$ emission reduction targets and increase import dependency. Instead, the operating lifetimes of the seven nuclear units should be extended.

In October 2009, the government received a further report from a commissioned panel[d]. This recommended a ten-year operating lifetime extension for the three oldest nuclear power reactors to 2025 and a 20-year lifetime extension for the other four. The government then agreed to postpone the phase-out by ten years, so that it would not begin before 2025[2]. This would allow the licensing of reactor lifetime extensions, and GDF Suez expected to invest about €800 million towards this. However, the operators agreed to contribute a special tax of €215-245 million per year over 2010-14 for the concession (0.5 Euro cents/kWh), and more thereafter. GDF Suez also agreed to subsidize renewables and demand-side management by paying at least €500 million for both, and it must maintain 13,000 jobs in energy efficiency and recycling. The Energy & Climate Change minister said that the delay in closure "would guarantee security of supply, limit the production of carbon dioxide and allow us to maintain prices that protect consumer purchasing power and the competitiveness of our companies."

However, an election in April 2010 occurred before the agreed proposals were passed by parliament and the nuclear phase-out law remained in place. In October 2011 several political parties negotiating to form a new government agreed that the 2003 nuclear phase-out law closing three reactors by 2015 and the others by 2025 should be implemented if adequate power could be secured from other sources and prices would not rise unduly.

In July 2012 Belgium's Council of Ministers announced that Doel 1&2 were to close in 2015 after 40 years of operation. However, Tihange 1, which also celebrated its 40th anniversary in 2015, was permitted to operate to 2025 to avoid the risk of blackouts. The other four Belgian reactors were not immediately affected by the decisions. This announcement reneged on "firm commitments" made between the Belgian state and GDF-Suez in a 2009 memorandum of understanding, and came immediately after the nuclear regulator FANC ruled favourably on a technical report from GDF Suez subsidiary Electrabel on its preparation for the ten-year extensions.

The government said that it had rewritten the 2003 law on nuclear energy so that its current stance could not be changed by decree, and therefore the timing of the phase-out "is now final". This certainty "should create a favourable investment climate which will allow us to gradually phase out nuclear power," it said. As the owner and operator of the country's nuclear power plants supplying half the country's electricity, Electrabel disagreed. It described the new shutdown dates as "blackmail", complaining that the attacks on the company and its 7000 staff were not justified. It also pointed out that operation of Tihange 1 to 2025 would require €500 million investment to satisfy safety regulators – an investment that would be hard to justify without "a vision and guarantees of profitability in ten years."

However in June 2015 parliament passed legislation to enable 10-year lifetime extensions for Doel 1&2 to 2025, thus overturning a major provision of the 2003 law (see above).

A report in October 2016 by PwC Enterprise Advisory considering scenarios to 2050 showed that Belgium would be unable to meet its climate goals while achieving long-term electricity price stability and security of supply without a significant contribution from nuclear energy beyond 2025.

In November 2017, three major Belgian business associations – Essenscia, Agoria, and Febeliec – said that shutting all nuclear plants by 2025 was not an affordable idea, and that the plan would boost carbon dioxide emissions and damage Belgian businesses.

At the end of March 2018 the government reaffirmed its phase-out policy and said that it would introduce capacity payments. Elia said that at least 3.6 GWe of new thermal capacity would be needed by the end of 2025, and in November 2020 Engie Electrabel said that it would build up to four natural gas plants by 2025.

In March 2022 Belgium delayed its plans to phase out nuclear energy by a decade. The prime minister said that two reactors (Doel 4 and Tihange 3) would continue operating to 2035 to "strengthen our county's independence from fossil fuels in a turbulent geopolitical environment." In June Engie said it was seeking financial aid from the government for the continued operation of the two reactors. An agreement between Engie Electrabel and the government on commencing studies on extended operation was announced in January 2022, outlining both parties' objective to restart Doel 4 and Tihange 3 in November 2026. Plans were finalized in December 2023, but the European Commission (EC) in July 2024 opened an investigation into whether the support for life

extension was in line with its rules on state aid. In February 2025 the EC approved the plan. The deal, which was closed in March 2025 results in:

- Equal ownership of Doel 4 and Tihange 3 by the Belgian state and Engie.
- Transfer of future nuclear waste liabilities from Engie to the Belgian state. Engie is to pay a fixed amount of €15 billion in two instalments, the first of which was made in March 2025, with the second to follow in November upon restart of the two reactors.

Earlier in February 2025 Belgium's new coalition government announced plans to operate the two units for an additional 10 years beyond 2035.

In May 2025 Belgium's parliament voted by a large majority to repeal the 2003 law for the phase-out of nuclear power.

## Nuclear taxes

In May 2010, CREG estimated the cost of producing electricity from Belgian nuclear power plants as 1.7-2.1 Euro cents/kWh, including fuel cycle, operating, depreciation, and provisions for decommissioning and waste management. This compared with the forward market price of 6 ¢/kWh and the market price for green energy certificates at 8.8-10.7 ¢/kWh.

In June 2013 Electrabel filed an appeal to Belgium's Constitutional Court against the €550 million ($730 million) annual federal tax on nuclear power generation. Earlier in 2012 the government passed laws doubling the size of the tax. As the dominant power generator and supplier, Electrabel bore the brunt of the tax – €479 million – while EDF-Luminus, Belgium's second-largest generator, has nuclear offtake rights and paid the remainder. GDF Suez said that raising the tax bill went against the protocol signed by the company and the federal government in 2009, which set out a special tax of €215-245 million for 2010-14.

In April 2014 the Court of First Instance in Brussels rejected Electrabel's claim it was entitled to an exceptional tax refund of about one billion euros over profits from its Belgian nuclear plants, which it had been paying 2008 to 2012 as a tax increment. The company appealed, saying the €479 million for 2012 corresponded to its entire nuclear profits, but that appeal was rejected in July

2014, with the court saying that the appeal was “unfounded”. Electrabel’s contribution of €422 million for 2013 was higher than its entire operational activities in Belgium. "This confiscatory fiscal pressure on Electrabel at a time when the company's economic situation has deteriorated resulted in losses for Electrabel in 2013 for the second year in succession." The company lodged an appeal against the 2013 contribution, but in September 2015 the Constitutional Court rejected this and said the tax was "legitimate".

In April 2015 CREG updated its 2010 calculations and said that the profit from all of Belgium's nuclear operators – Electrabel, EDF Luminus and EDF Belgium – totalled some €435 million in 2014. CREG reported: "The profits derived from nuclear activities are the difference between revenues and costs. The 'nuclear rent' can then be calculated by subtracting a fair remuneration of the invested capital." Electrabel noted that the nuclear contribution amounted to €479 million in 2014 – some €44 million more than CREG’s calculation of profits. "The nuclear operators are thus paying to the state more than they earn from this nuclear activity," and CREG's latest calculation clearly shows that the 'nuclear contribution' imposed by the government on nuclear operators since 2008 is confiscatory, it said.

In July 2015 Electrabel agreed to pay €130 million in 2016 as the federal nuclear power ‘contribution’ or tax, substantially less than previously intended (and less than €479 million in 2014). This is alongside a fee for lifetime extension of Doel 1&2 – see section above. From 2017 a formula was applied, with a minimum of €150 million per year to 2019.

Earlier in mid-2013 the government approved an energy plan which would subsidize gas-fired generation and offshore wind capacity with taxes from nuclear power. Investors had been deterred from investing in planned 800 MWe of gas-fired plant by the relatively low cost of nuclear power and the grid priority of renewables input.

In October 2013 the government reached agreement with Electrabel and EDF so that they would get a fixed price of €41.8 per MWh produced at Tihange 1 over 2015 to 2025. If the market price exceeds the fee, the owners will get 30% of the excess proceeds, the balance to the state.

## Fuel cycle

The fuel cycle for Belgian nuclear plants is managed by Synatom (*Société Belge des Combustibles Nucléaires*), a wholly-owned subsidiary of Electrabel (although the government holds a golden share with special rights).

In 2008, Synatom announced a strategic partnership with Powertech for development of Powertech's US uranium projects in the US states of Colorado, South Dakota and Wyoming. In November 2013, Synatom sold its stake (then 15.3%) in Powertech to Azarga Resources[3].

Fuel fabrication is by Areva's (now Framatome's) 700 t/yr FBFC (Franco-Belgian Fuel Fabrication) International Dessel plant. The facility includes a 200 t/yr MOX fuel fabrication building. The 35 t/yr Belgonucléaire[e] MOX plant, also at Dessel, closed in 2006 and is being decommissioned. Mixed oxide (MOX) fuel has been used in Belgium since 1995 and the policy is to use all recovered plutonium from reprocessing for MOX fuel fabrication as quickly as possible.

The Eurochemic reprocessing plant at Dessel was built as a joint facility with 12 other OECD states. It operated 1966 to 1974 and is being dismantled by Belgoprocess[f].

With Eurochemic closed, Synatom then contracted with Cogema (now Orano) for reprocessing used fuel at La Hague from 1976. However, political attitudes towards reprocessing changed in the late 1980s and, in 1993, parliament decided to suspend the 1990 reprocessing contract and to not negotiate any new reprocessing contracts. The once-through fuel cycle option was also given the same level of priority as reprocessing.

## Radioactive waste management

The National Agency for Radioactive Waste and Enriched Fissile Materials – ONDRAF/NIRAS – is responsible for the management of all radioactive materials in the country, including transport, treatment, conditioning, storage and disposal. Its main facility is at the Mol-Dessel site, run by its subsidiary Belgoprocess. Its costs are passed onto the producers of radioactive waste, notably the power companies. Utilities pay a levy on each kWh of electricity sold, which goes into a decommissioning and waste management fund, managed by Synatom.

A number of shipments of vitrified high-level waste from reprocessed Belgian fuel have taken place from La Hague in France. The wastes are stored at Dessel. However, following the

1993 political decision to suspend reprocessing activities, used fuel is currently stored at the nuclear power plants. Most of the country's high-level waste originates from the Eurochemic reprocessing plant.

In June 2006, the government decided that low-level and short-lived intermediate-level wastes would be disposed of in a surface repository at Dessel[i]. The municipality of Mol had also been considered and expressed a willingness for the facility to be there. Construction of the surface repository began in September 2025. Civil engineering work will be conducted by the NucleusSafe partnership, with construction company Denys responsible for electromechanical aspects. The facility will consist of several concrete bunkers that will house large concrete vaults in which short-lived low- and intermediate-level waste will have been encapsulated with mortar. 28,831 vaults are planned, spread across two zones: 20 bunkers in the first and 14 in the second.

Research on deep geological disposal of long-lived intermediate-level and high-level wastes is underway and focused on the clays at Mol. In 1980, construction of the Hades (*High-activity disposal experimental site*) underground research laboratory 225 m deep in the Boom clay commenced[g]. The management and operation of Hades is carried out by the Economic Interest Grouping EURIDICE (*European underground research infrastructure for disposal of radioactive waste in a clay environment*), which was set up by ONDRAF/NIRAS and the Belgian Nuclear Research Centre (SCK.CEN). The main objective of EURIDICE is to carry out the PRACLAY (*Preliminary demonstration test for clay disposal of highly radioactive waste*) project, which aims to demonstrate the feasibility of disposing of radioactive waste in deep clay layers[h].

In 2016 Synatom contracted with Germany's *Gesellschaft für Nuklear-Service mbH* (GNS) to provide 30 CASTOR casks from 2021 for transport and storage of used fuel at Doel and Tihange.

# Decommissioning

Decommissioning activities are now well advanced at several early nuclear facilities. These include:

- The 40 MWt BR3 prototype PWR reactor which was started up in 1962 and closed down in 1987 is being decommissioned by SCK.CEN.

- The Franco-Belgian Chooz A in France was closed in 1991 and is being decommissioned by EDF.
- The Eurochemic reprocessing plant ended reprocessing activities in 1974 and decommissioning by Belgoprocess began in 1989.
- Decommissioning of Belgonucleaire's MOX fuel fabrication plant.

SCK.CEN has built up expertise in decommissioning through the BR3 decommissioning project. The 11 MWe BR3 was the first PWR in Europe and also, in 1987, became the first PWR in Europe to be shut down. In 1989, it was selected as a European Commission Framework Programme pilot project for the demonstration of decommissioning of PWR plants.

## Research and development

R&D is mainly carried out by the Belgian Nuclear Research Centre (SCK/CEN) at Mol. It runs the 100 MWt BR2 high-flux tank-type research reactor which started up in 1961 but has been modified since, and two small units. BR2 produces about 16% of the world's Mo-99/Tc-99m for medical use[1]. Before they are administered to patients, isotopes from BR2 are treated with a chemical process at the Institut National des Radioéléments. The residues left behind during this process still contain a number of substances that can be recovered. In February 2023 RECUMO, a facility to recover radioactive residues from the production of medical radioisotopes, began construction in Mol. The facility is scheduled to begin operation in 2026 and would process the current residues and those resulting from isotope production until 2038.

In May 2025 SCK/CEN and the national nuclear regulator began a consultation on a lead-cooled SMR reactor design. It is part of the EU-SMR-LFR project, in which SCK/CEN collaborates with partners in Italy and Romania.  In the long term, the project provides for the construction of two precursors: the first on the SCK-CEN site in Mol, followed by a second in Pitesti, Romania.

In 1967, Belgium joined with Germany and Holland to build a 300 MWe prototype fast breeder reactor at Kalkar in Germany. The plant was completed but never run. Belgium also participated in France's Superphénix fast reactor, which operated between 1985 and 1996.

## Myrrha

SCK.CEN is planning to build the Myrrha (Multipurpose Hybrid Research Reactor for High-tech Applications) research reactor at Mol. Initially it will be a 57 MWt accelerator-driven system (ADS), consisting of a proton accelerator delivering a 600 MeV, 2.5 mA (or 350 MeV, 5 mA) proton beam to a liquid lead-bismuth (Pb-Bi) spallation target that in turn couples to a Pb-Bi cooled, subcritical fast nuclear core[i]. As an ADS it will be used to prove that technology and to study transmutation of long-lived radionuclides in nuclear waste. Later on it is intended to be run as a critical fast neutron facility, decoupling the accelerator and removing the spallation loop from the reactor core. Then Myrrha will be used for fuel research, for materials research for Generation IV reactors, and for the production of radioisotopes and doped silicon (an essential component of high-grade electronic circuits).

It is envisaged as a partnership of Belgium, the European Union, the European Investment Bank and others, with 70% of the funding from EU countries. Following an independent international evaluation[4], Belgium approved its 40% share of the funding for construction costs in March 2010. The project is expected to cost €1.6 billion. In 2010, construction was expected to start in 2015, with operation from about 2024, but progress has been slower than anticipated[5]. A reduced-power demonstration model, Guinevere, became operational at Mol in March 2010.

In October 2010, SCK.CEN signed two international agreements to collaborate on the Myrrha project. The first is with the China Academy of Sciences, since China sees Myrrha as a way forward in treating nuclear waste. The second is with Kazatomprom and the Kazakh National Nuclear Centre (NNC). SCK.CEN said that these agreements "constitute the first step towards these Asian countries' probable entry into the Myrrha consortium"[6] that will fund construction and operation of the facility. In October 2013 SCK-CEN awarded a €24 million contract for engineering design to a consortium led by Areva TA, and including Ansaldo Nucleare of Italy and Empresarios Agrupados of Spain. In September 2018, Belgium's Council of Ministers approved €558 million in funding for Myrrha. The funding "will help carry out the first important part of Myrrha: building the first part of its particle accelerator and target facilities," known as the Minerva installation. A groundbreaking ceremony for the construction of Minerva took place in June 2024. The funding would be provided over the period 2019-2038, firstly to cover by 2026 the studies and the development of the Minerva

installation, and secondly the operating costs after 2027. Construction of the Myrrha reactor itself is expected to begin in 2026, with full operation from 2034.

## Public opinion

Recent polling suggests public views about the use of nuclear energy are generally positive. For more information see page on Nuclear Energy and Public Opinion.

## Regulation, safety and non-proliferation

The Federal Agency for Nuclear Control (FANC) operates under the Minister of Interior. Its focus is on radiation safety, both occupational and public, under a 1994 Act which replaced a 1958 Act.

The private nuclear safety company Association Vinçotte Nuclear, formerly AIB-Vinçotte Nuclear (AVN), carried out technical assessment activities on behalf of FANC, and in April 2008, AVN was transferred into Bel V, which is the technical safety organization within FANC.

### Non-proliferation

Belgium is a party to the Nuclear Non-Proliferation Treaty (NPT) since 1975 as a non-nuclear weapons state. It is member of the Nuclear Suppliers Group. The Additional Protocol in relation to its safeguards agreements with the International Atomic Energy Agency was signed in 1998 and came into force in 2004.

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## Notes & references

### Notes

a. In 1905, a group of electricity companies combined to form Scheldt Electricity Company, which in 1956 merged with three other companies to form Ebes. In 1990, Ebes merged its operations with those of Intercom and Unerg, forming Electrabel.

Meanwhile, in 1986, engineering consultancy Electrobel (note spelling: this is a separate organization to Electrabel) merged with engineering services provider Tractionel to form Tractebel. In the mid-1990s, Tractebel acquired a 40% stake in Electrabel.

A majority stake in Tractebel was held by Société Générale de Belgique (SGB), which in 1988 was taken over by Compagnie Financière de Suez, a predecessor company to Suez Lyonnaise des Eaux. By 1996 SGB had built up a stake of over 65% in Tractebel but this decreased to 50.3% a few months later during the merger of Tractebel and Powerfin (another SGB subsidiary). In 1997, SGB owner Compagnie Financière de Suez merged with Lyonnaise des Eaux-Dumez to form Suez Lyonnaise des Eaux.

In 2003, Tractebel merged with SGB to form Suez-Tractebel, a wholly-owned subsidiary of Suez. By the end of 2003, Suez had increased its holding in Electrabel to 50.1% and, by July 2007, had acquired all the shares in the company. In the meantime, Electrabel operated as a subsidiary of Suez-Tractebel.

GDF Suez was formed by the merger of Gaz de France and Suez in July 2008 and changed its name to Engie in April 2015. [Back]

b. While a series of mergers of private utilities by 1980 resulted in the three companies Ebes, Intercom and Unerg – which later merged to form Electrabel (see Note a) – all the public utilities regrouped to form SPE in 1979.

In 2005, UK-based utility Centrica formed a joint venture with Gaz de France (GDF) to acquire a 51% stake in SPE. Later in 2005, SPE merged with Luminus, the second largest energy supplier in Belgium to form SPE-Luminus. In 2008, Centrica acquired GDF's share of SPE-Luminus when, as part of the conditions attached to GDF's merger with Suez, GDF was required to dispose of its stake in SPE. In 2009, Centrica sold its stake in SPE-Luminus to EDF as part of a deal to buy a 20% stake in British Energy from EDF. SPE-Luminus was renamed EDF Luminus in 2011.

EDF was holding a 63.5% stake in EDF-Luminus through EDF Belgium and Segebel, with the remaining 36.5% owned by Publilec, Dexia Bank, Socofe, VEH, Ethias, Publilum and ALG. In 2010, ALG was integrated into Tecteo Services Group (now Nethys). Also in 2010, Dexia sold its shareholding, and Publilum and VEH sold half their stakes. In 2015 EDF agreed to increase its stake to 68.63%, by buying out Publilum and VEH. The remainder of EDF Luminus is owned by Ethias, Nethys, Publilec and Socofe.

EDF Luminus owns 10% of nuclear capacity in Belgium and has rights to the output from another 5% of capacity. [Back]

c. The *Commission on Energy 2030* was set up at the end of 2005 at the request of Marc Verwilghen, Federal Minister of Energy. Its final report[7] was published in June 2007. [Back]

d. The GEMIX report[8] published in October 2009 was commissioned by Belgian Minister for Climate and Energy Paul Magnette. See also: Nuclear needed in Belgium, World Nuclear News (2 October 2009). [Back]

e. Established in 1957, Belgonucléaire operated the P0 mixed oxide fuel plant at Dessel from its startup in 1973 (with commercial production in 1986) through to its closure in 2006. Belgonucléaire is half-owned by the Belgian government through the Belgian Nuclear Research Centre (SCK.CEN) and the other half of its ownership now comes under GDF Suez, following acquisitions of Tractebel and Electrabel. Belgonucléaire jointly owns the Belgatom architect and consulting engineering company founded in 1978. The majority owner in Belgatom was Tractebel but, following the restructuring under GDF Suez, Belgatom generally trades under the Tractebel Engineering name. [Back]

f. With a capacity of 60 tonnes per year, the Eurochemic reprocessing plant was uncompetitive and it was decided to temporarily close the facility in 1975. Eventually, the Belgian government agreed to take it over to supply domestic needs only and, in 1984, Belgoprocess was founded to manage the plant. However, in 1985, the decision was taken to not restart the plant and Belgoprocess was transferred to the Belgian National Agency for Radioactive Waste and Irradiated Fissile Materials (ONDRAF/NIRAS). [Back]

g. Construction on the Hades underground research laboratory was started by the Belgian Nuclear Research Centre (SCK.CEN) in 1980 by sinking a shaft through the aquifer sands into the Boom Clay, up to 230 m deep. (The Boom Clay layer at Mol is 100 m thick and starts at a depth of 180 m.) The major phases in the development of Hades are the construction of: the first access shaft (1980-1982); the underground research laboratory (1982-1983); the experimental shaft and gallery (1983-1984); the extension of the URL with the test drift (1987); the second access shaft (1997-1999); the connecting gallery (2001-2002); and the PRACLAY gallery (2006-2007, see Note k below).

ONDRAF/NIRAS was created in 1981 and became operational in 1983. Since then, ONDRAF/NIRAS has gradually taken over the

responsibility for nuclear waste disposal with SCK.CEN remaining the main research partner. In 1995, ONDRAF/NIRAS and SCK.CEN set up the Economic Interest Group (EIG) PRACLAY (*Preliminary demonstration test for clay disposal of highly radioactive waste*), which in 1997 became EIG EURIDICE (*European underground research infrastructure for disposal of radioactive waste In a clay environment*), to manage and operate Hades. [Back]

h. Information on the PRACLAY project is available on the EURIDICE website (www.euridice.be). [Back]

i. See the MYHHRA section of the SCK.CEN website. [Back]

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